

MBEYA UNIVERSITY OF SCIENCE AND TECHNOLOGY
COLLEGE OF INFORMATION AND COMMUNICATION TECHNOLOGY

Test 01: ET 8208 Object-Oriented Programming

Date 2nd May 2024

Instructions

- Answer all questions
- The time allocated for this test is sixty (60) minutes

Question One

What do you understand by the following terms in object-oriented programming

- Class
- Methods Overloading
- Polymorphism
- Encapsulation
- Access modifiers
- Static members

[3 Marks]

keyword = define the accessibility of a class and its member.

Question two

- With at least one example from each, differentiate Procedural programming from Object-oriented Programming

[2 Marks]

- Write a flow chart for a method to convert decimal numbers to binary numbers

[4 Marks]

Question three

- Using JAVA running codes, differentiate parameterized from non-parameterized constructors.

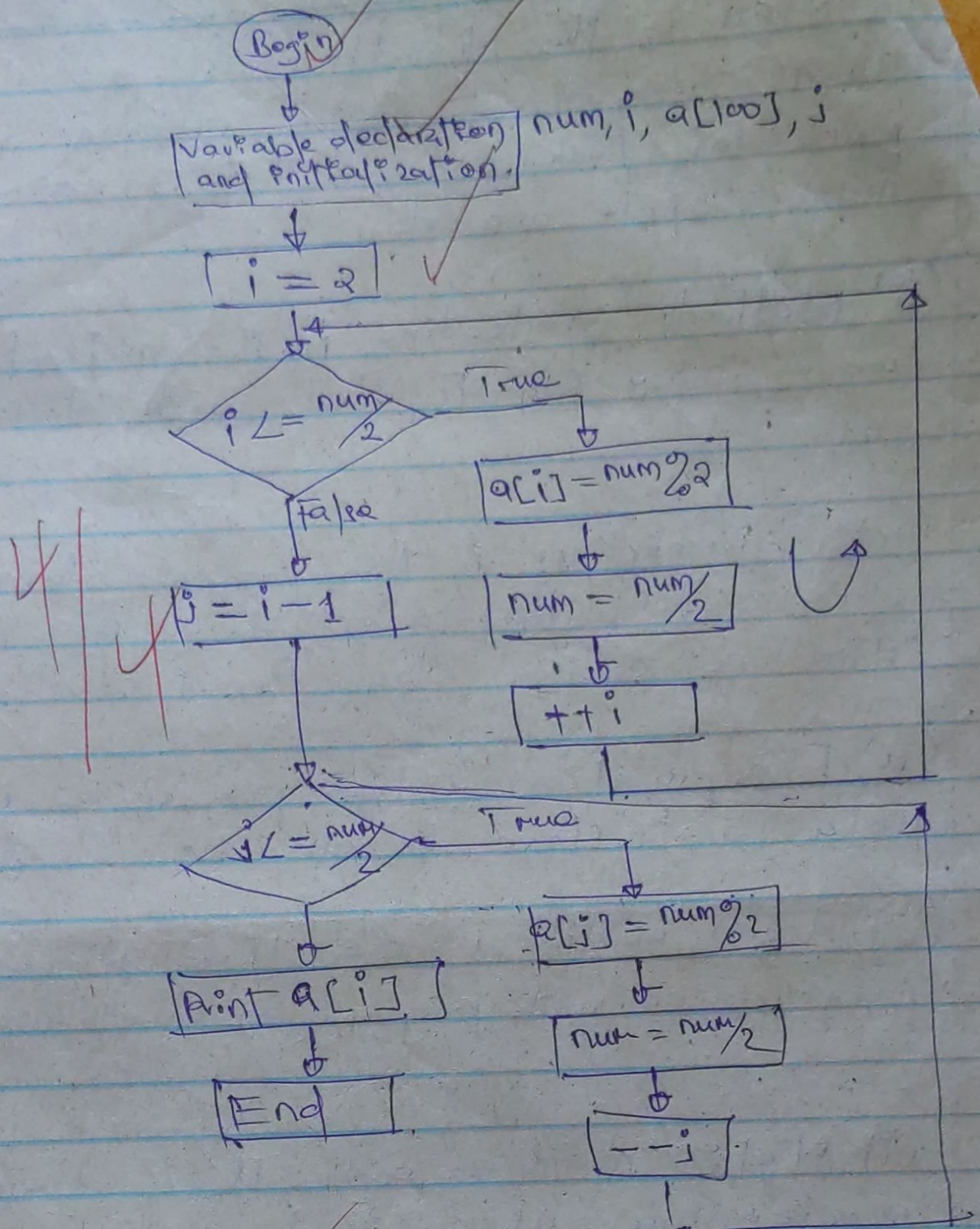
[3 Marks]

- What are the outputs of the JAVA program below?

[3 Marks]

```
import java.io.*;
class Marks{
int m1;
int m2;
}
class MarksDemo
{
public static void main(String args[])
{ //first object
Marks obj1 = new Marks();
obj1.m1 = 50;
obj1.m2 = 80;
//second object
Marks obj2 = new Marks();
obj2.m1 = 80;
obj2.m2 = 60;
//displaying marks for first object
System.out.println("Marks for first object:");
System.out.println(obj1.m1);
System.out.println(obj1.m2);
//displaying marks for second object
System.out.println("Marks for second object:");
System.out.println(obj2.m1);
System.out.println(obj2.m2);
}
}
```


Q. (ii).



(i) Procedural programming	Object-oriented programming.
- Data and logic are separated.	- Data and logic are not separated.
- It include C language	- It include C++, JAVA, Python languages.

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1 Program of finding area of triangle.

Parameterized Constructor.

```
class Triangle {  
    double height;  
    double base;
```

```
    Triangle(double h, double b) {  
        height = h;  
        base = b;  
    }
```

```
    area ( ) {  
        return 0.5 * height * base;  
    }  
}
```

```
class DemoTriangle {
```

```
    public static void main (String args[]) {  
        double a;
```

```
        Triangle myTriangle1 = new Triangle (6, 10);
```

```
        a = myTriangle1.area ( );
```

```
        System.out.println ("The area is" + a )
```

```
    }  
}
```

2 Non parameterized Constructor

```
class Triangle {  
    double height;  
    double base;
```

```
    Triangle ( ) {  
        height = 6;  
        base = 10;  
    }
```

```
    area ( ) {  
        return 0.5 * height * base;
```


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3. (i) class DemoTriangle {
public static void main (String args[]) {
double a;
Triangle myTriangle1 = new Triangle ();
a = myTriangle1.area ();
System.out.println ("The area is " + a);
}
}

(ii). Output.

Marks for first object: 50 80 ✓

Marks for second object: 80 60 ✓

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